

1D CNN — Architecture Variant Comparison

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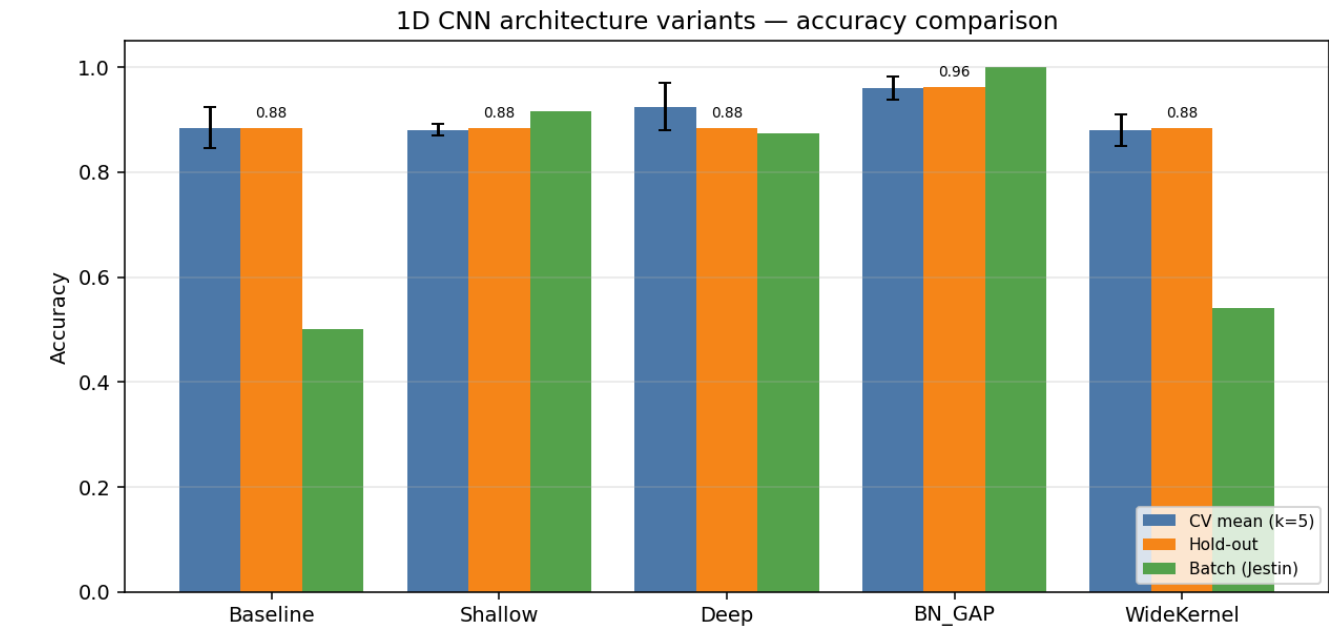
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Total trials	252 (train pool 226, hold-out 26)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

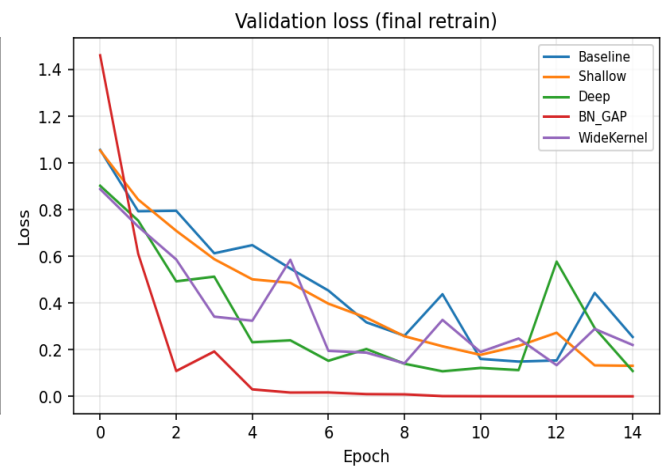
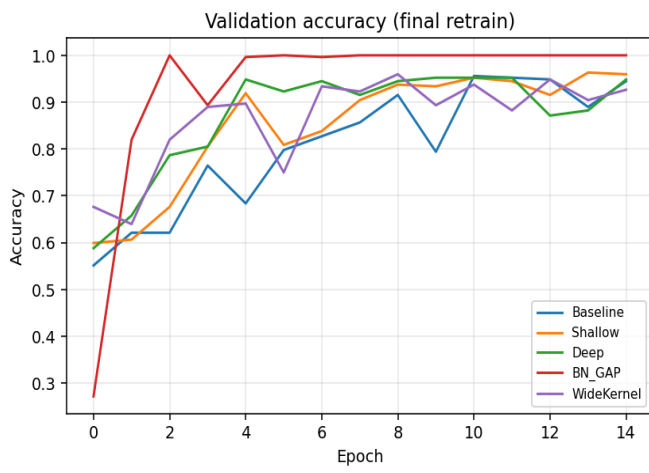
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.8847	0.0386	0.8846	0.7821	0.5000	12/24
Shallow	72,487	0.8805	0.0113	0.8846	0.6180	0.9167	22/24
Deep	204,007	0.9251	0.0446	0.8846	0.8049	0.8750	21/24
BN_GAP	57,319	0.9603	0.0214	0.9615	0.2179	1.0000	24/24
WideKernel	131,687	0.8806	0.0299	0.8846	0.7635	0.5417	13/24

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.8847 ± 0.0386
Hold-out test acc / loss	0.8846 · 0.7821
Batch-test acc	0.5000 (12/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1080	46	0.9348	0.9348	0.9348
2	1086	45	0.8222	0.8889	0.8222
3	1086	45	0.8667	0.9111	0.8667
4	1086	45	0.9111	0.9111	0.9111
5	1086	45	0.8889	0.9556	0.8889

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.250	0.400	4
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.800	1.000	0.889	4
Double_cmere	0.800	1.000	0.889	4
Drum_Roll	0.750	1.000	0.857	3
macro avg	0.907	0.893	0.862	26
weighted avg	0.910	0.885	0.857	26

Batch test per class (Jestin)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	3	5	0.600
Double_Wiggle	2	4	0.500
Double_cmere	2	5	0.400
Overall	12	24	0.500

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8805 ± 0.0113
Hold-out test acc / loss	0.8846 · 0.6180
Batch-test acc	0.9167 (22/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1080	46	0.8913	0.8913	0.8913
2	1086	45	0.8667	0.8889	0.8667
3	1086	45	0.8889	0.9111	0.8889
4	1086	45	0.8889	0.8889	0.8889
5	1086	45	0.8667	0.9111	0.8667

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.250	0.400	4
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.667	1.000	0.800	4
Double_cmere	0.800	1.000	0.889	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.924	0.893	0.870	26
weighted avg	0.918	0.885	0.860	26

Batch test per class (Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	3	5	0.600
Double_Wiggle	4	4	1.000
Double_cmere	5	5	1.000
Overall	22	24	0.917

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9251 ± 0.0446
Hold-out test acc / loss	0.8846 · 0.8049
Batch-test acc	0.8750 (21/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1080	46	0.8478	0.9130	0.8478
2	1086	45	0.9556	1.0000	0.9556
3	1086	45	0.9333	0.9556	0.9333
4	1086	45	0.9778	0.9778	0.9778
5	1086	45	0.9111	0.9778	0.9111

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.250	0.400	4
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.667	1.000	0.800	4
Double_cmere	0.800	1.000	0.889	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.924	0.893	0.870	26
weighted avg	0.918	0.885	0.860	26

Batch test per class (Jestin)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	3	5	0.600
Double_Wiggle	4	4	1.000
Double_cmere	5	5	1.000
Overall	21	24	0.875

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9603 ± 0.0214
Hold-out test acc / loss	0.9615 · 0.2179
Batch-test acc	1.0000 (24/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1080	46	0.9348	0.9565	0.9348
2	1086	45	0.9778	1.0000	0.9778
3	1086	45	0.9333	0.9778	0.9333
4	1086	45	0.9778	1.0000	0.9778
5	1086	45	0.9778	1.0000	0.9778

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.750	0.857	4
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	4
Double_cmere	0.800	1.000	0.889	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.971	0.964	0.964	26
weighted avg	0.969	0.962	0.961	26

Batch test per class (Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	4	4	1.000
Double_cmere	5	5	1.000
Overall	24	24	1.000

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8806 ± 0.0299
Hold-out test acc / loss	0.8846 · 0.7635
Batch-test acc	0.5417 (13/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1080	46	0.8696	0.8913	0.8696
2	1086	45	0.8444	0.9111	0.8444
3	1086	45	0.9333	0.9333	0.9333
4	1086	45	0.8889	0.9333	0.8889
5	1086	45	0.8667	0.9111	0.8667

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.250	0.400	4
Double_Pistol_Recoil	1.000	1.000	1.000	3
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.667	1.000	0.800	4
Double_cmere	0.800	1.000	0.889	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.924	0.893	0.870	26
weighted avg	0.918	0.885	0.860	26

Batch test per class (Jestin)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	1	5	0.200
Double_Wiggle	4	4	1.000
Double_cmere	3	5	0.600
Overall	13	24	0.542